

KKT KOLBE HERMES RGBW Hood

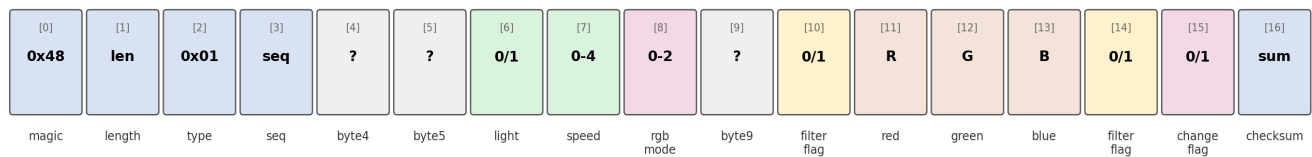
Hekr Protocol Reference — Address Space Map

This document maps every address space in the raw-hex protocol used to control the cooker hood over its Hekr/Wisen cloud connection (hub.hekreu.me:83), reverse-engineered through packet capture, systematic command sweeping, official firmware SDK cross-reference, and community protocol documentation. It covers two distinct address spaces: the **byte layout** within each frame, and the **command ID space** that selects what a frame does.

1. Status Frame Byte Layout

Sent by the device to the cloud (and relayed to Home Assistant) roughly every second, and immediately after any state change. Frame type byte is always 0x01 for this direction.

Status Frame Byte Layout (device -> cloud, e.g. 481101E501000000020000398C69000171)



Offset	Field	Description
0	Magic byte	Constant 0x48. Marks the start of every frame, both directions.
1	Length	Total frame length in bytes, including this byte and the final checksum.
2	Frame type	0x01 for a status report. (0x02 is used for the command direction — see Section 2.)
3	Sequence	Rolling counter incremented by the device on every frame it sends. Not used for authentication.
4	byte4	Unidentified. Observed as 0x01 in nearly every capture; role unknown.
5	byte5	Unidentified. Occasionally changes alongside light/speed during panel use; exact role unclear.
6	Light (white)	0 = off, 1 = on. The white light channel, independent of the RGB colour. Settable via Command ID 0x03.
7	Fan speed	0 = off, 1–4 = speed levels. Settable via Command ID 0x04.
8	RGB mode flag	0 = never engaged (seen only on first-ever connection). 1 = off, colour value remembered. 2 = on, colour genuinely displayed. Fully solved 2026-08-13: directly and reliably set via cmdId 0x07's mode byte alone (mode=0x02 for 2, mode=0x00 for 1) — no fan, reset, or "Light" command (cmdId 0x03) involvement needed. Confirmed accurate against the physical hood at every transition (see Section 4).
9	byte9	Unidentified. Always 0x00 in every capture so far.
10	Filter reminder flag	0 = no warning, 1 = filter needs cleaning. Confirmed 2026-08-13 against a real, organically-triggered filter warning. Mirrors byte 14 exactly — both flip together. Previously listed as unidentified/always-0x00; only ever differs from 0 once the panel's actual filter indicator is lit.
11	Red	0–255. Confirmed by cycling all 9 physical panel colour presets and cross-referencing each against this byte.
12	Green	0–255. Confirmed as above.
13	Blue	0–255. Confirmed as above.

Offset	Field	Description
14	Filter reminder flag	Mirrors byte 10 (see above). Reason for the duplication unknown — possibly a redundancy/checksum-adjacent design choice, possibly two related-but-distinct conditions that happen to always coincide in testing so far.
15	Change flag	Transient “just changed” pulse: briefly 1 right after any state change, then reverts to 0 on the next heartbeat. Ignore when reading steady-state values.
16	Checksum	Sum of every preceding byte in the frame, modulo 256 (simple byte-sum, no CRC polynomial). Confirmed identical across three independent open-source implementations of this protocol.

2. Command Frame Shapes (app/bridge → device)

Commands sent to the device follow the same magic-byte/length/type/sequence header, but with frame type 0x02, and a payload shape that depends on the Command ID. Four shapes are confirmed:

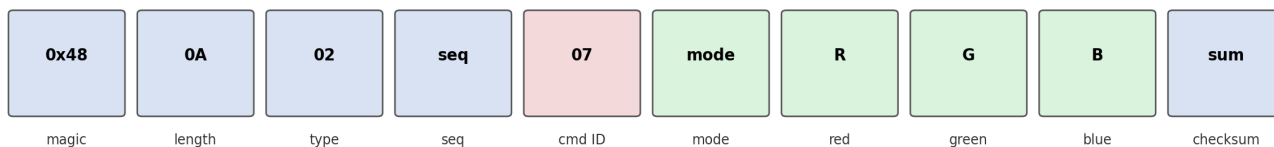
Simple command (Power 0x02, Light 0x03, Speed 0x04, Timer 0x05)

example: 480702D70301 2C -> Light ON



Colour command (cmdId 0x07 only)

example: 480A02E80702398C6973 -> set colour to #398C69, RGB on



Timer command (cmdId 0x05)

example: 480702D80501 2F -> auto-off timer, duration unit 1



Clock-set command (cmdId 0x08) — confirmed 2026-08-11

example: 48090204080F270095 -> set clock to 15:39:00



Note the colour command's extra **mode** byte before R/G/B — without it, the device silently misinterprets the frame; this was the first key discovery that unlocked reliable colour control at all. The mode byte's own value turned out to matter just as much, and took longer to pin down correctly: **mode=0x02** genuinely displays the given colour, and **mode=0x00** genuinely turns RGB output off (indicator included) — a clean, symmetric, completely fan-free on/off pair, with no other command involved (see Section 4).

3. Command ID Address Space

Command ID occupies byte offset 4 of a command frame and selects which function the payload targets. Every value from 0x02 through 0x14 has been tested; results below.

ID	Function	Status	Notes
0x02	Power	Working	Value byte: 0/1. Fan-specific in practice: only has any observable effect when the fan is currently running (confirmed during the RGB investigation, Section 4) — sending it while the fan is already off does nothing at all, regardless of light or RGB state. The Home Assistant Power entity reflects fan speed only for this reason.
0x03	Light (white)	Working	White light only, independent of RGB. Value byte: 0/1.
0x04	Fan speed	Working	Value byte: 0–4.

ID	Function	Status	Notes
0x05	Auto-off timer	Working	Value byte sets countdown duration (units unconfirmed, roughly seconds-to-minutes). Panel-cancellable. On completion, only resets its own duration byte (offset 9) — confirmed to have zero effect on the RGB mode flag (offset 8) or any other field. A value of 0x02 produces a ~90-second countdown; retested cleanly on 2026-08-11 after an earlier ambiguous log (from 2026-07-28) raised a false alarm suggesting it might toggle the RGB flag directly — that turned out to be coincidental timing with a manual panel press, not a real effect.
0x06	Filter/clean reminder reset	Working, confirmed	Value byte 0x00. Solved 2026-08-13: earlier extensive testing (bare, 1-byte, extended 2-byte payloads) consistently found no response, which turned out to be because a filter-reset command has no observable effect while no filter warning is active. Retested against a real, organically-triggered warning and confirmed immediately — response arrived in under one second and cleared both status bytes 10 and 14, plus the physical panel icon.
0x07	RGB colour	Fully solved, 2026-08-13	Requires 4-byte payload: mode + R + G + B. Full mode semantics finally confirmed after a couple of false starts along the way: mode=0x02 genuinely drives the RGB hardware, displaying the given colour immediately (status byte 8 -> 2). mode=0x00 genuinely turns the RGB output off — confirmed dark, including the physical panel indicator (status byte 8 -> 1) — while remembering the given colour, when sent directly, with no reset or fan involvement needed either way. mode=0x02/mode=0x00 form a clean, symmetric on/off pair. mode=0x03 is accepted and behaves identically to 0x02. mode=0x01 is never acknowledged by the device, in any context, across dozens of tests. Two earlier hypotheses about this mode byte both turned out to be incomplete before this was pinned down precisely — see Section 4.
0x08	Clock (time-of-day)	Working, confirmed	Two distinct uses confirmed. (1) Simple 1-byte values accidentally adjust the wall-clock display (early discovery, required a manual reset). (2) Proper 3-byte payload — [hour, minute, second], with a genuinely computed checksum, not the hardcoded 0x00 a reference implementation used — reliably sets the clock to an exact time. Confirmed working 2026-08-11 and built into the add-on as a “Sync Clock” button. The clock is never reported back in the status frame — this command is fire-and-forget, verified visually only.
0x09	RGB on/off (per community docs)	Documented, not functional	Identified via an independent open-source adapter's protocol table (cmdId 9 = “RGB_Sw”). That project's own author marked it “not implemented” after being unable to get a response. Exhaustively retested on this hood: bare, 1-byte, extended 2-byte, the reference adapter's exact literal format, a 3-byte colour-shaped payload, and the full 4-byte [mode,R,G,B] shape that unlocked cmdId 0x07 — 15 distinct variants total, tested with the RGB light confirmed genuinely on beforehand (to rule out silence-because-nothing-to-turn-off). No response in any case.
0x0A–0x0D	—	No response	Tested bare and with 1-byte values (0/1). Silent every time.
0x0E–0x14	—	No response	Tested bare, with 1-byte values, and (0x0E–0x10 only) with a 3-byte RGB-shaped payload. Silent every time.
various	Frame-type byte variants	No effect	Tested 0x00, 0x01, 0x03, 0x04 in place of the usual 0x02 frame-type byte, paired with cmdId 0x07. No response from any.

4. How This Maps to the Home Assistant Add-on

The hekr_bridge add-on builds and parses these exact frames. Practical behaviour, given everything above:

- **Power / Light / Fan Speed** entities map directly to Command IDs 0x02/0x03/0x04 — fully reliable. Power reflects fan speed only (not light state) — see the Command ID 0x02 note above.
- **RGB Light** entity: genuine, fan-free on/off fully solved 2026-08-13, on the third attempt at characterising this correctly (see the note below). Final design: turning on (with or without a new colour) sends **mode=0x02** with the target R/G/B; turning off sends **mode=0x00** with the last known R/G/B. Both are sent directly, with no reset, no fan involvement, and no black-colour workaround needed at any point. **Light (cmdId 0x03) is not involved in RGB on/off at all** — it was a red herring during the investigation; it only ever controls the independent white channel.
- **On/off status** shown in Home Assistant reads directly from status byte 8 (the RGB flag), which is fully reliable under this design — it also correctly reflects colour changes made from the physical panel, not just from Home Assistant.
- **Note on how this was pinned down:** two earlier hypotheses about the mode byte were tried and believed correct before this one - first that mode=0x01 was the off-flag (never acknowledged by the device at all), then that mode=0x00 plus a Light on/off toggle gave clean fan-free control (the status flag genuinely toggled cleanly, but the colour was never actually visible on that path - confirmed only by explicitly checking the physical hood, which the earlier conclusion had skipped). The working design above was confirmed by directly observing the hood at every step, including two full repeated on/off cycles sent with no reset in between.
- **Sync Clock** button (added 2026-08-11) uses Command ID 0x08 to set the hood's clock to the add-on's current system time, or an explicit override time if published to the same topic — useful if the add-on container's timezone doesn't match the hood's physical location.
- **Filter Needs Cleaning** binary sensor and **Reset Filter Reminder** button (added 2026-08-13) read and clear status bytes 10/14 via Command ID 0x06 — confirmed working against a real filter warning, closing out what had been an open investigation since the project began.
- **RGB “last known colour”** is read directly from the hood's own most recent status frame rather than cached separately by the add-on — the device reports its stored R/G/B on every status update regardless of on/off state, so this is simpler, always accurate, survives add-on restarts with no file needed, and stays in sync with colour changes made from the physical panel too.

Sources cross-referenced during this investigation: direct packet capture and command sweeping against this device; Hekr's official HEKR-MCU-SDK reference firmware; the ioBroker.hekr community adapter; the hekrapi-python protocol library; and the decompiled Wisen Android application.

Revised 2026-08-13 (latest): Power entity fixed to reflect fan speed only, not light state (matches confirmed cmdId 0x02 behaviour). RGB last-known-colour now read directly from the hood's own status reports instead of cached separately. Revised 2026-08-13 (final): genuine RGB off fully and correctly solved. mode=0x02 (on) and mode=0x00 (off) are a clean, directly-usable, completely fan-free pair — confirmed against the physical hood across repeated cycles. Two earlier hypotheses this same day (mode=0x01 as the off-flag; then mode=0x00 plus a Light on/off toggle) were each believed correct before being disproven by actually checking the hood rather than trusting the status byte alone. No project limitations remain outstanding. Revised 2026-08-13 (earlier): filter/clean reminder fully solved (indicator bytes and reset command both confirmed against a real warning). Revised 2026-08-11: added confirmed clock-set command, retested cmdId 0x05/0x06 findings.